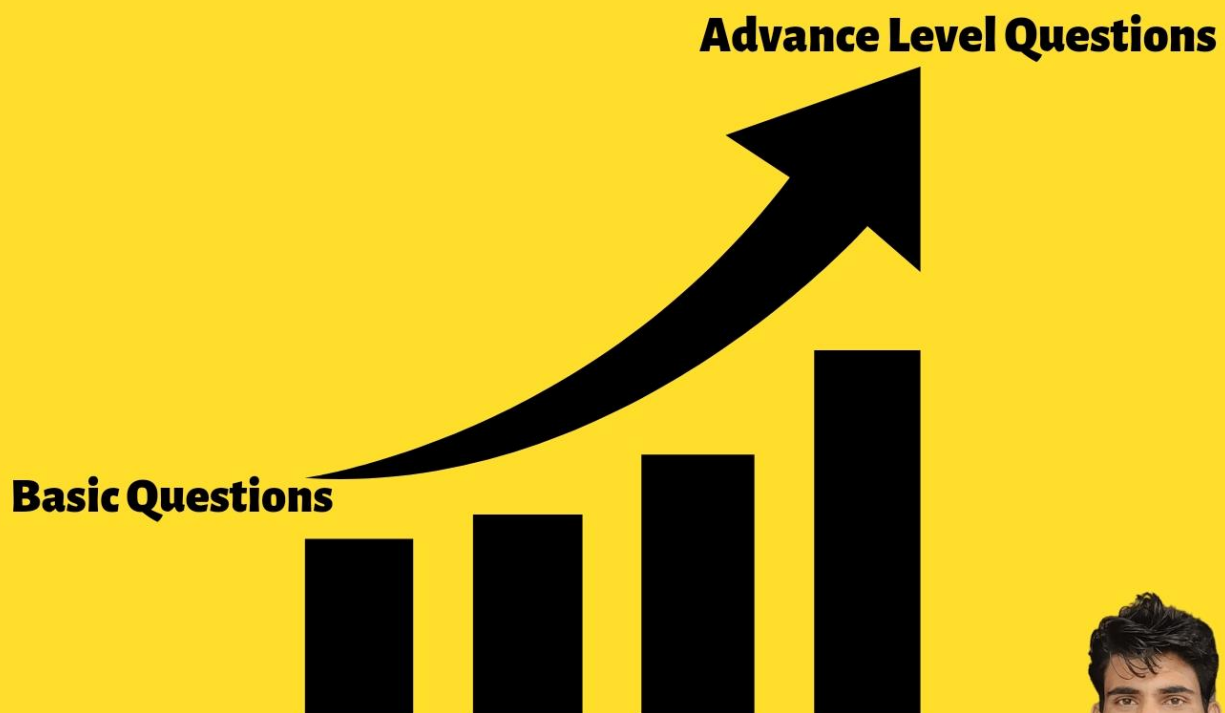


MATHS

SURDS & INDUCES

ALL TYPE WISE QUESTIONS



Maths with Gajanand



Surds & Indices

Rules of Indices :-

1. $a \times a \times a \times a \dots n \text{ times} = a^n$
2. $a^m \times a^n = a^{m+n}$
3. $\frac{a^m}{a^n} = a^{m-n}$ if $m > n$
 $= \frac{1}{a^{n-m}}$ if $n > m$
 $= 1$ if $m = n$
4. $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
5. $(a^m)^n = a^{m \times n} = a^{n \times m} = (a^n)^m$
6. $a^{-n} = \frac{1}{a^n}$
7. $(ab)^n = a^n \times b^n$
8. If $a^m = a^n \Rightarrow \boxed{m = n}$
9. If $a^m = b^m \Rightarrow \boxed{a = b}$
10. $(a^m)^n \neq a^{m^n}$
 $(3^2)^4 \neq 3^{2^4}$
 $3^8 \neq 3^{16}$

$$* a^0 = 1$$

$$* (-1)^n \begin{cases} \rightarrow +1 & \text{if } n = \text{even} \\ \rightarrow -1 & \text{if } n = \text{odd} \end{cases}$$

$$* \left(\frac{a}{b}\right)^m = \left(\frac{b}{a}\right)^{-m}$$

$$* a^n = \frac{1}{a^{-n}}$$

[1] If $(2^3)^2 = 4^x$ then 3^x is equal to:

Soln:- $2^6 = 2^{2x}$

$$6 = 2x \Rightarrow \boxed{x = 3} \text{ Ans}$$

[2] If $3^{x+y} = 81$ and $81^{x-y} = 3$, then the value of x is:

Soln:- $3^{x+y} = 3^4$, $3^{4(x-y)} = 3^1$

$$x+y = 4, \quad x-y = \frac{1}{4}$$

$$x = \frac{4 + \frac{1}{4}}{2} \Rightarrow \boxed{\frac{17}{8}} \text{ Ans}$$

[3] If $m^n = 169$, What is the value of $(m+1)^{(n-1)}$?

Soln:- $m^n = (13)^2$

$$m = 13$$

$$n = 2$$

$$(m+1)^{(n-1)}$$

$$\Rightarrow (14)^1 = \boxed{14} \text{ Ans}$$

[4] $(64)^{-\frac{2}{3}} \times \left(\frac{1}{4}\right)^{-2}$ is equal to :

Soln :- $\frac{1}{(64)^{2/3}} \times (4)^2$

$\Rightarrow \frac{1}{16} \times 16 = \boxed{1}$ Ans

[5] If $(125)^{2/3} \times (625)^{-1/4} = 5^x$, then the value of x is

Soln :- $25 \times 5^{-1} = 5^x$

$5^{2-1} = 5^x \quad \boxed{x=1}$ Ans

[6] If $x^{x\sqrt{x}} = (x\sqrt{x})^x$, then $x = ?$

Soln :- $x^{x^{3/2}} = (x^{3/2})^x$

$x^{x^{3/2}} = x^{3/2 x}$

$x^{3/2} = \frac{3}{2}x$

$x^{1/2} = \frac{3}{2} \Rightarrow \boxed{x = \frac{9}{4}}$ Ans

[7] $\left\{ (-2)^{(-2)} \right\}^{(-2)} = ?$

Soln :- $\frac{1}{\left\{ (-2)^{(-2)} \right\}^2} \Rightarrow \frac{1}{(-2)^{-4}} \Rightarrow (-2)^4 = \boxed{16}$ Ans

[8] $256^{-[4^{3/2}]} = ?$

Soln :- $256^{-[4^{3/2}]} \Rightarrow 256^{-[8]}$

$\Rightarrow \frac{1}{(256)^8} \Rightarrow \frac{1}{(2^8)^8} \Rightarrow \boxed{\frac{1}{2}}$ Ans

9] The value of $\frac{(243)^{\frac{n}{5}} \times 3^{2n+1}}{9^n \times 3^{n-1}}$ is:

Soln:-

$$\Rightarrow \frac{(3^5)^{\frac{n}{5}} \times 3^{2n+1}}{3^{2n} \times 3^{n-1}}$$

$$\Rightarrow \frac{3^n \times 3^{2n+1}}{3^{2n} \times 3^{n-1}} \Rightarrow \frac{3^{n+2n+1}}{3^{2n+n-1}}$$

$$\Rightarrow \frac{3^{3n+1}}{3^{3n-1}}$$

$$\Rightarrow 3^{(3n+1)-(3n-1)}$$

$$\Rightarrow 3^2 = \boxed{9} \text{ Ans}$$

10] If $\frac{3^{a+3} \times 4^{a+6} \times 25^{a+1}}{27^{a-1} \times 8^{a-2} \times 125^{a+4}} = \frac{4}{15^{26}}$, then $\sqrt{a+9} = ??$

Soln:- $\frac{3^{a+3} \times 2^{2a+12} \times 5^{2a+2}}{3^{3a-3} \times 2^{3a-6} \times 5^{3a+4}} = \frac{4}{15^{26}}$

$$\frac{2^{(2a+12)-(3a-6)}}{3^{(3a-3)-(a+3)} \times 5^{(3a+4)-(2a+2)}} = \frac{2^2}{3^{26} \times 5^{26}}$$

$$\frac{2^{-a+18}}{3^{2a-6} \times 5^{a+10}} = \frac{2^2}{3^{26} \times 5^{26}}$$

Comparison both side $2^{-a+18} = 2^2$

$$-a+18 = 2$$

$$\boxed{a = 16}$$

$$\sqrt{a+9} = \sqrt{25} = \boxed{5} \text{ Ans}$$

11 If $\left[\left\{ \left(\frac{2}{3} \right)^3 \right\}^{(2x+3)} \right]^{-\frac{3}{4}} = \left[\left\{ \left(\frac{2}{3} \right)^{\frac{2}{3}} \right\}^{(3x+7)} \right]^{-\frac{6}{5}}$

Soln:-

then the value of $\sqrt{2-42x}$ is:

$$\Rightarrow 3 \times (2x+3) \times -\frac{3}{4} = \frac{2}{3} \times (3x+7) \times -\frac{6}{5}$$

$$\Rightarrow 90x + 135 = 48x + 112$$

$$\Rightarrow 42x = -23$$

$$\sqrt{2-42x} = \sqrt{2+23} = \underline{\underline{5}} \text{ Ans}$$

12 $\frac{9^n \times 3^2 \times (3^{-\frac{n}{2}})^{-2} - (27)^n}{3^{3m} \times 2^3} = \frac{1}{729}$. then $m-n = ?$

Soln:- $\frac{3^n \times 3^2 \times 3^n - 3^{3n}}{3^{3m} \times 2^3} = \frac{1}{3^6}$

$$\Rightarrow \frac{3^{3n} \times 3^2 - 3^{3n}}{3^{3m} \times 2^3} = \frac{1}{3^6}$$

$$\Rightarrow \frac{3^{3n} (3^2 - 1)}{3^{3m} \times 2^3} = \frac{1}{3^6}$$

$$\Rightarrow \frac{1}{3^{3m-3n}} = \frac{1}{3^6}$$

$$3m-3n = 6$$

$$m-n = \underline{\underline{2}} \text{ Ans}$$

Laws of Surds

$$1. \quad \sqrt[n]{a} = a^{\frac{1}{n}}$$

$$2. \quad \sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b} = a^{\frac{1}{n}} \times b^{\frac{1}{n}} = (ab)^{\frac{1}{n}}$$

$$3. \quad (\sqrt[n]{a})^n = (a^{\frac{1}{n}})^n = a$$

$$4. \quad \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \left(\frac{a}{b}\right)^{\frac{1}{n}}$$

$$5. \quad (\sqrt[n]{a})^m = a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

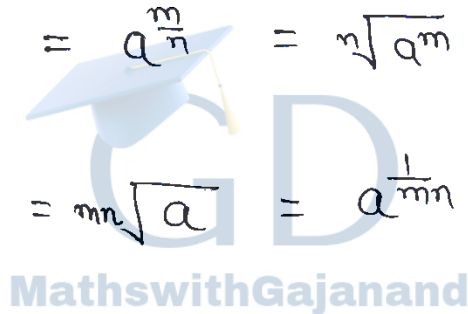
$$6. \quad \sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a} = a^{\frac{1}{mn}} = \sqrt[n]{\sqrt[m]{a}}$$

$$7. \quad \sqrt[w]{z \sqrt[y]{x \sqrt[a]{a}}} = a^{\frac{1}{xyzw}} = \sqrt[w]{xyz \sqrt[a]{a}}$$

$$8. \quad \left(z \sqrt{ \left(y \sqrt{ \left(x \sqrt{a} \right)^m } \right)^n } \right)^o = a^{\frac{mno}{xyz}}$$

$$9. \quad a^{\frac{m}{n}} = (a^m)^{\frac{1}{n}}$$

$$10. \quad \text{if } x^n = a \Rightarrow x = a^{\frac{1}{n}}$$



13] Simplified form of $\left[\left(\sqrt[5]{x^{-3/5}} \right)^{-5/3} \right]^{-5}$ is:

Soln :- $\Rightarrow x^{\frac{-3/5 \times -5/3 \times -5}{5}}$
 $\Rightarrow x^{-1} = \boxed{\frac{1}{x}}$ Ans

14] Simplify:

$$\left[\sqrt[12]{\left(\sqrt[6]{\left(\sqrt[3]{5^4} \right)^8} \right)^9} \right]^{18} = ?$$

Soln :-

$$5^{\frac{4 \times 8 \times 9 \times 18}{3 \times 6 \times 12}} \Rightarrow 5^{24} \text{ Ans}$$

15]

$$\left[8 - \left(\frac{4^{\frac{3}{2}} \sqrt{2 \cdot 2^2}}{2 \cdot \sqrt{2^{-2}}} \right)^{\frac{1}{2}} \right] \text{ is equal to:}$$

Soln :- $8 - \left(\frac{2^{3/2} \sqrt{2^3}}{2 \times \sqrt{\frac{1}{4}}} \right)^{1/2}$
 $\Rightarrow 8 - \left(\frac{2^{3/2} \cdot 2^{3/2}}{2 \times \frac{1}{2}} \right)^{1/2}$
 $\Rightarrow 8 - (2^6)^{1/2}$
 $\Rightarrow 8 - 2^3$
 $\Rightarrow 8 - 8$
 $\Rightarrow \boxed{0} \text{ Ans}$

Type- Simplification

Type-1 $\sqrt{a + \sqrt{a + \sqrt{a + \dots \infty}}} = \frac{1 + \sqrt{4a+1}}{2} = x$

$$\sqrt{a - \sqrt{a - \sqrt{a - \dots \infty}}} = \frac{\sqrt{4a+1} - 1}{2} = y$$

*
$$\begin{cases} x - y = 1 \\ xy = a \end{cases}$$
$$x + y = \sqrt{4a+1}$$

Trick Take two factors of a whose difference is 1

$\begin{matrix} & x & y \\ & \text{(Larger)} & \text{(Smaller)} \end{matrix}$

Ex:-

$$\sqrt{30 + \sqrt{30 + \sqrt{30 + \dots}}} = ?$$

factor:- 5 6

→ 6 Ans (if sign is $\oplus$ ive then Larger factor)

Ex:-

$$\sqrt{42 - \sqrt{42 - \sqrt{42 - \dots}}} = ?$$

6 7
Ans

(If sign is $\ominus$ ve then Answer will be smaller factor)

16] What is the value of $2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}} = ?$

Soln:- $2 + 2 = \boxed{4}$ Ans

17] $\sqrt{31 + \sqrt{31 + \sqrt{31 + \dots}}} = ?$

- (a) $5\sqrt{5} - 1.5$ (b) $2.5\sqrt{5} + 0.5$ (c) $\frac{5\sqrt{5} - 1}{2}$ (d) $\frac{2\sqrt{31} + 1}{2}$

Soln:- $\frac{\sqrt{4 \times 31 + 1} + 1}{2} = \frac{5\sqrt{5} + 1}{2} \Rightarrow \boxed{2.5\sqrt{5} + 0.5}$ Ans

18] $\sqrt{19 - \sqrt{19 - \sqrt{19 - \dots}}} = ?$

- (a) $\frac{\sqrt{77} - 1}{2}$ (b) $\frac{\sqrt{19} + 3}{2}$ (c) $\frac{\sqrt{77} + 1}{2}$ (d) Between 4 & 5

Soln:- $\frac{\sqrt{4 \times 19 + 1} - 1}{2} \Rightarrow \boxed{\frac{\sqrt{77} - 1}{2}}$ Ans

19] $\sqrt{14 + \sqrt{14 + \sqrt{14 + \sqrt{14 + \dots}}}} = ?$ lies between

- (a) 4 and 4.5 (b) 4.5 and 5 (c) 3 and 4 (d) None

Soln:- $\frac{\sqrt{4 \times 14 + 1} + 1}{2} = \frac{\sqrt{57} + 1}{2}$

$\frac{7+1}{2} < \frac{8+1}{2}$

$\Rightarrow \boxed{4 \text{ \& } 4.5}$ Ans

$\sqrt{57}$
 $\swarrow \quad \searrow$
 $\sqrt{49} \quad \sqrt{64}$
 7 8

20 $m = \sqrt{5 + \sqrt{5 + \sqrt{5} \dots}}$, $n = \sqrt{5 - \sqrt{5 - \sqrt{5} \dots}}$

then among the following the relation between m & n hold is:

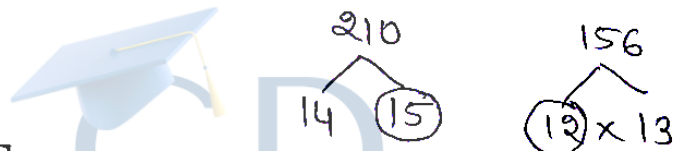
- (a) $m - n + 1 = 0$ (b) $m + n + 1 = 0$ (c) $m + n - 1 = 0$ (d) $m - n + 1 = 0$

Soln:- $m - n = 1$

$m - n - 1 = 0$ Ans

21
$$\frac{\sqrt{210 + \sqrt{210 + \sqrt{210 + \dots}}}}{\sqrt{156 - \sqrt{156 - \sqrt{156 - \dots}}}} = ?$$

Soln $\Rightarrow \frac{15}{12} = \frac{5}{4}$
 $= 1.25$ Ans



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22 The value of $p + \sqrt{p^2 + \sqrt{p^4 + \sqrt{p^8 + \sqrt{p^{16} + \dots}}}}$

- (a) $p\left(\frac{\sqrt{5}+2}{2}\right)$ (b) $p\left(\frac{3+\sqrt{5}}{2}\right)$ (c) $p\left(\frac{p}{1+\sqrt{p}}\right)$ (d) $p\left(\frac{\sqrt{5}+1}{2}\right)$

Soln:- Take $p = 1$

$\therefore 1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + \dots}}}$

$\Rightarrow 1 + \frac{\sqrt{5}+1}{2}$

$\Rightarrow \frac{\sqrt{5}+3}{2}$ option (b) ✓

Type-2

$$x = \sqrt{a + \sqrt{a - \sqrt{a + \dots \infty}}} \Rightarrow \frac{\sqrt{4a-3} + 1}{2}$$

$$y = \sqrt{a - \sqrt{a + \sqrt{a - \dots \infty}}} \Rightarrow \frac{\sqrt{4a-3} - 1}{2}$$

$$* \boxed{x-y=1}$$

$$\boxed{xy = 2(a-1)}$$

Q3 let $x = \sqrt{4 + \sqrt{4 - \sqrt{4 + \sqrt{4 - \dots \infty}}}}$ then x equals,

Soln:-

$$\frac{\sqrt{4 \times 4 - 3} + 1}{2} \Rightarrow \boxed{\frac{\sqrt{13} + 1}{2}}$$

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Q4

$$x = \sqrt{7 - \sqrt{7 + \sqrt{7 - \sqrt{7 + \dots \infty}}}} = ?$$

Soln:-

$$\frac{\sqrt{4 \times 7 - 3} - 1}{2} \Rightarrow \frac{\sqrt{25} - 1}{2}$$

$$\Rightarrow \frac{5-1}{2}$$

$$\Rightarrow \boxed{2} \text{ Ans}$$

Type-2

$$x = \sqrt{a+b} \sqrt{a+b} \sqrt{a+b} \dots \infty = \frac{\sqrt{4a+b^2} + b}{2}$$

$$y = \sqrt{a-b} \sqrt{a-b} \sqrt{a-b} \dots \infty = \frac{\sqrt{4a+b^2} - b}{2}$$

* $\begin{cases} x-y=b \\ xy=a \end{cases}$

$x+y = \frac{\sqrt{4a+b^2}}{2}$

Ex:- Find $\sqrt{35+2} \sqrt{35+2} \sqrt{35+2} \sqrt{35+2} \dots \infty = ?$

Sol:- **m-I** Basic

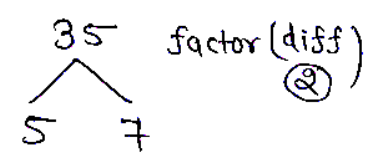
$$\begin{aligned} x &= \sqrt{35+2x} \\ x^2 &= 35+2x \\ x^2-2x-35 &= 0 \\ (x-7)(x+5) &= 0 \\ x &= 7, -5 \times \\ \boxed{x=7} \end{aligned}$$

m-II

$$\begin{aligned} &\frac{\sqrt{4 \times 35 + 4 + 2}}{2} \\ &\Rightarrow \frac{\sqrt{144+2}}{2} \\ &\Rightarrow \frac{14}{2} \\ &\Rightarrow \boxed{7} \text{ Ans} \end{aligned}$$

m-III

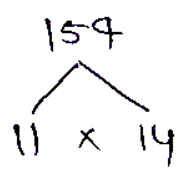
Trick



Sign +ve $\Rightarrow$ greater
 $\Rightarrow \boxed{7}$ Ans

Q5 Find $\sqrt{154-3} \sqrt{154-3} \sqrt{154-3} \sqrt{154-3} \dots \infty = ?$

Sol:- Trick



Sign:- -ve so smaller factor
 Ans = $\boxed{11}$

Type-4

$$\sqrt{a+b}\sqrt{a-b}\sqrt{a+\dots} = \frac{\sqrt{4a-3b^2} + b}{2}$$

$$\sqrt{a-b}\sqrt{a+b}\sqrt{a-\dots} = \frac{\sqrt{4a-3b^2} - b}{2}$$

Q6 $\sqrt{7+2\sqrt{7-2\sqrt{7+\dots}}} = ?$

Soln:- $\frac{\sqrt{4 \times 7 - 3 \times 2^2} + 2}{2} \Rightarrow \frac{\sqrt{16} + 2}{2} = \frac{6}{2} = \boxed{3}$ Ans

Q7 $\sqrt{10-3\sqrt{10+3\sqrt{10-\dots}}} = ?$

Soln:- $\frac{\sqrt{4 \times 10 - 3 \times 9} - 3}{2} \Rightarrow \frac{\sqrt{13} + 3}{2}$ Ans

Q8 $\sqrt[3]{210 + 3\sqrt[3]{210 + 3\sqrt[3]{210 + \dots}}} = ?$

Soln:- **m-I** Basic

$$x = \sqrt[3]{210 + x}$$

$$x^3 = 210 + x$$

$$x^3 - x = 210$$

$$x(x^2 - 1) = 210$$

$$x(x-1)(x+1) = 5 \times 6 \times 7$$

$\boxed{x = 6}$ Ans

m-II Trick

210 (factor in diff. 1)
 $\begin{array}{c} 210 \\ \swarrow \downarrow \searrow \\ 5 \times 6 \times 7 \end{array}$

middle no. is Ans

$\boxed{6}$ Ans

Type-5 $\sqrt{a\sqrt{a\sqrt{a\sqrt{a\cdots\infty}}}} = \boxed{a}$

$$\Rightarrow \sqrt{a\sqrt{a\sqrt{a\cdots n \text{ times}}}} = a^{1-\frac{1}{2^n}} = a^{\frac{2^n-1}{2^n}}$$

Proof :- $a^{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots n \text{ times}}$ G.P

$$\Rightarrow a^{\frac{\frac{1}{2}[1-(\frac{1}{2})^n]}{1-\frac{1}{2}}} \Rightarrow a^{1-\frac{1}{2^n}} \quad \left[T_n = \frac{a[1-r^n]}{1-r} \quad \text{if } r < 1 \right]$$

$$\Rightarrow \sqrt[n]{a\sqrt[n]{a\sqrt[n]{a\cdots\infty}}} = \sqrt[n]{a}$$

$$\Rightarrow \sqrt[n]{a \div \sqrt[n]{a \div \sqrt[n]{a\cdots\infty}}} = \sqrt[n+1]{a}$$

29 $\sqrt{5\sqrt{5\sqrt{5\sqrt{5\cdots\infty}}}} \Rightarrow \boxed{5}$ Ans

30 $\sqrt{12\sqrt{12\sqrt{12\sqrt{12\sqrt{12\sqrt{12}}}}}} = ?$

Soln :- (a) $12^{\frac{32}{31}}$ (b) $12^{\frac{64}{63}}$ (c) $12^{\frac{31}{32}}$ (d) $12^{\frac{63}{64}}$

$$12^{\frac{2^6-1}{2^6}}$$

$$\Rightarrow \boxed{12^{\frac{63}{64}}} \text{ Ans}$$

31

$$\sqrt[3]{64 \sqrt[3]{64 \sqrt[3]{64} \dots \infty}} = ?$$

Soln:-

$$(3-1)\sqrt[3]{64} \Rightarrow 2\sqrt[3]{64} = \boxed{8} \text{ Ans}$$

32

$$\sqrt{27 \div \sqrt{27 \div \sqrt{27 \div \dots \infty}}} = ?$$

Soln:-

$$(2+1)\sqrt{27} = 3\sqrt{27} = \boxed{3}$$

m-II Basic:-

$$x = \sqrt{\frac{27}{x}} \Rightarrow x^2 = \frac{27}{x} \\ \Rightarrow x^3 = 27 \Rightarrow \boxed{x = 3} \text{ Ans}$$

33

Find $\sqrt{2 \times \sqrt[3]{4 \times \sqrt{2 \times \sqrt[3]{4 \times \dots \infty}}}} = ?$

(a) $\sqrt{2}$

(b) 2

(c) 4

(d) $4\sqrt{2}$

Soln:- m-I Basic:

$$x = \sqrt{2 \times \sqrt[3]{4 \times \sqrt{2 \times \sqrt[3]{4 \times \dots}}}}$$

$$(x)^2 = (\sqrt{2 \times \sqrt[3]{4 \times \sqrt{2 \times \dots}}})^2$$

$$x^2 = 2 \sqrt[3]{4 \times \sqrt{2 \times \sqrt[3]{4 \times \dots}}}$$

Cube

$$x^6 = 8 \times 4 \sqrt{2 \times \sqrt[3]{4 \times \dots}}$$

$$x^6 = 8 \times 4 \times x^2$$

$$x^5 = 32$$

$$x = \sqrt[5]{32} = \boxed{2} \text{ Ans}$$

m-II

$$2 \times 3 - 1 \sqrt{2^3 \times 4}$$

$$\Rightarrow \sqrt[5]{32}$$

$$\Rightarrow \boxed{2} \text{ Ans}$$

[34] If $x = \sqrt{a\sqrt[3]{b}\sqrt{a\sqrt[3]{b}\dots\infty}}$, then value of x is:

- (a) $\sqrt{ab^3}$ (b) $\sqrt[3]{a^5b}$ (c) $\sqrt[3]{a^3b}$ (d) $\sqrt[5]{a^3b}$

Solⁿ

$$x = \sqrt{a\sqrt[3]{bx}}$$

$$x^2 = a\sqrt[3]{bx}$$

$$x^6 = a^3bx$$

$$x^5 = a^3b$$

$$x = \boxed{\sqrt[5]{a^3b}} \text{ Ans}$$

[35] If $x = \sqrt{a\sqrt[3]{ab}\sqrt{a\sqrt[3]{ab}\dots\infty}}$, then the value of x is:

Solⁿ:-

$$x = \sqrt{a\sqrt[3]{abx}}$$

square:- $x^2 = a\sqrt[3]{abx}$

cube:- $x^6 = a^3abx$

$$x^5 = a^4b$$

$$x = \sqrt[5]{a^4b}$$

[36] $\sqrt[m]{a\sqrt[n]{b}\sqrt[m]{a\sqrt[n]{b}\dots\infty}}$ can be written as:

- (a) $\sqrt[mn]{a^n b}$ (b) $\sqrt[mn]{ab}$ (c) $\sqrt[mn-1]{bna}$ (d) $\sqrt[mn]{a^n b}$

Solⁿ

$$x = \sqrt[m]{a\sqrt[n]{bx}}$$

$$x^m = a\sqrt[n]{bx}$$

$$(x^m)^n = a^n bx$$

$$x^{mn-1} = a^n b$$

$$x = \boxed{\sqrt[mn]{a^n b}} \text{ Ans}$$

37 Solve $\sqrt{21 + 3\sqrt{59 + \sqrt{16 + 3\sqrt{729 + \sqrt{49}}}}} = ?$

(a) 4

(b) 5

(c) 6

(d) 7

Soln:-

$$\begin{aligned} &\sqrt{21 + 3\sqrt{59 + \sqrt{16 + 3\sqrt{729 + \sqrt{49}}}}} \\ &\quad \quad \quad \downarrow \\ &\quad \quad \quad \underbrace{3\sqrt{729} = 9} \\ &\quad \quad \quad \underbrace{\sqrt{16 + 9} = 5} \\ &\quad \quad \quad \underbrace{3\sqrt{59 + 5} = 4} \\ &\quad \quad \underbrace{\sqrt{21 + 4} = 5} \text{ Ans} \end{aligned}$$

38 If $x^m = \sqrt[14]{x \sqrt{x \sqrt{x}}}$, then what is the value of m ?

(a) $\frac{1}{8}$

(b) $\frac{1}{4}$

(c) $\frac{3}{4}$

(d) $\frac{7}{4}$

Soln

$$x^m = x^{\frac{1}{14}} \cdot x^{\frac{1}{28}} \cdot x^{\frac{1}{56}}$$

$$= x^{\frac{1}{14} + \frac{1}{28} + \frac{1}{56}}$$

$$= x^{\frac{1+2+4}{56}}$$

$$x^m = x^{\frac{1}{8}}$$

$$\boxed{m = \frac{1}{8}} \text{ Ans}$$

39 If

$$\sqrt{x} \sqrt{x} \sqrt{x} \sqrt{x} \dots = \frac{1}{2} \text{ then } x = ?$$

(a) $\frac{1}{8}$

(b) $\frac{1}{4}$

(c) $\frac{1}{16}$

(d) $\frac{1}{32}$

Soln:-

$$(\sqrt{x})^{\frac{1}{2}} = \frac{1}{2}$$

$$x^{\frac{1}{4}} = \frac{1}{2}$$

$$x = \frac{1}{2^4} \Rightarrow \boxed{\frac{1}{16}} \text{ Ans}$$

40 If

$$x \sqrt{x} \sqrt{x} \sqrt{x} \dots = \frac{1}{36} \text{ then } x = ?$$

(a) 6^{-12}

(b) 6^{-18}

(c) 6^{-8}

(d) 6^{-16}

Soln:-

$$(\sqrt{x} \times \sqrt{x})^{\sqrt{x} \sqrt{x} \dots} = \frac{1}{36}$$

$$\Rightarrow \sqrt{x} \sqrt{x} \sqrt{x} \dots \times \sqrt{x} \sqrt{x} \sqrt{x} \dots = \frac{1}{6} \times \frac{1}{6}$$

$$\text{So } \sqrt{x} \sqrt{x} \sqrt{x} \dots = \frac{1}{6}$$

$$(\sqrt{x})^{\frac{1}{6}} = \frac{1}{6}$$

$$x^{\frac{1}{12}} = \frac{1}{6}$$

$$x = \left(\frac{1}{6}\right)^{12}$$

$$x = \boxed{6^{-12}} \text{ Ans}$$

Type - Square-root

$$\Rightarrow \sqrt{7 + 4\sqrt{3}} \Rightarrow \sqrt{\underset{\substack{\uparrow \\ a^2}}{(2)^2} + \underset{\substack{\uparrow \\ b^2}}{(\sqrt{3})^2} + \underset{\substack{\uparrow \\ 2ab}}{2 \times 2 \times \sqrt{3}}}$$

$$\Rightarrow \sqrt{(2 + \sqrt{3})^2}$$

$$\Rightarrow \boxed{2 + \sqrt{3}}$$

$$\Rightarrow \sqrt{61 + 28\sqrt{3}} \Rightarrow \sqrt{(7)^2 + (2\sqrt{3})^2 + 2 \times 7 \times 2\sqrt{3}}$$
$$\Rightarrow \sqrt{(7 + 2\sqrt{3})^2}$$
$$\Rightarrow \boxed{7 + 2\sqrt{3}}$$

$$\Rightarrow \sqrt{28 + 10\sqrt{3}} \Rightarrow \begin{array}{c} \text{short trick} \\ 10\sqrt{3} \\ \swarrow \searrow \\ 2 \times 5\sqrt{3} \\ \uparrow \quad \uparrow \\ a \quad b \end{array} \Rightarrow \boxed{\frac{a+b}{2}} = \boxed{5 + \sqrt{3}}$$

$$\Rightarrow \sqrt{17 + 12\sqrt{2}} \Rightarrow \begin{array}{c} 12\sqrt{2} \\ \swarrow \searrow \\ 2 \times 6\sqrt{2} \\ \swarrow \searrow \\ 3 \times 2\sqrt{2} \\ 3^2 + (2\sqrt{2})^2 = 17 \end{array} \Rightarrow \boxed{3 + 2\sqrt{2}}$$

$$\Rightarrow a^2 + b^2 - 2ab = (a-b)^2 = (b-a)^2$$

$$\sqrt{a^2 + b^2 - 2ab} \begin{cases} \rightarrow a-b & \text{if } a > b \\ \rightarrow b-a & \text{if } b > a \end{cases}$$

$$\begin{aligned} * \quad \sqrt{43 - 24\sqrt{3}} &\Rightarrow \begin{array}{c} 24\sqrt{3} \\ \swarrow \searrow \\ \textcircled{2} \times 12\sqrt{3} \\ \swarrow \searrow \\ 4 \quad 3\sqrt{3} \\ \swarrow \searrow \\ 16 + 27 = 43 \end{array} \Rightarrow \boxed{(3\sqrt{3} - 4)} \end{aligned}$$

$$\begin{aligned} \Rightarrow \sqrt{139 - 80\sqrt{3}} &\Rightarrow \begin{array}{c} 80\sqrt{3} \\ \swarrow \searrow \\ \textcircled{5} \times 16\sqrt{3} \\ \swarrow \searrow \\ 5\sqrt{3} \quad 8 \\ \swarrow \searrow \\ 75 + 64 = 139 \end{array} \Rightarrow \boxed{5\sqrt{3} - 8} \text{ Ans} \end{aligned}$$

$$\begin{aligned} \Rightarrow \sqrt{11 - 3\sqrt{8}} &\Rightarrow \begin{array}{c} 3\sqrt{8} \\ \Downarrow \\ 3 \times 2\sqrt{2} \\ \swarrow \searrow \\ \textcircled{3} \times 2\sqrt{2} \\ \swarrow \searrow \\ 3 \quad \sqrt{2} \\ \swarrow \searrow \\ 9 + 2 = 11 \end{array} \Rightarrow \boxed{3 - \sqrt{2}} \end{aligned}$$

$$\Rightarrow \sqrt{6 - \sqrt{35}} = ?$$

$$\begin{aligned} \underline{\text{Soln}}: \quad \sqrt{\frac{12 - 2\sqrt{35}}{2}} &\Rightarrow \begin{array}{c} 2\sqrt{35} \\ \swarrow \searrow \\ \textcircled{2} \times \sqrt{35} \\ \swarrow \searrow \\ \sqrt{5} \quad \sqrt{7} \\ \swarrow \searrow \\ 5 + 7 = 12 \end{array} \Rightarrow \boxed{\frac{\sqrt{7} - \sqrt{5}}{\sqrt{2}}} \text{ Ans} \end{aligned}$$

* If $x = 7 + 4\sqrt{3}$ then $\frac{1}{x} = ?$

$$\begin{aligned}\Rightarrow \frac{1}{x} &= \frac{1}{7+4\sqrt{3}} \times \frac{7-4\sqrt{3}}{7-4\sqrt{3}} \\&= \frac{7-4\sqrt{3}}{7^2 - (4\sqrt{3})^2} \\&= \frac{7-4\sqrt{3}}{49-48} \\&= \frac{7-4\sqrt{3}}{1} \leftarrow \text{Sign change} \\&\quad \leftarrow \text{difference of square,}\end{aligned}$$

Ex:- $x = 5 + 2\sqrt{6}$ then $\frac{1}{x} = ?$

Soln:- $\frac{1}{x} = \frac{5-2\sqrt{6}}{5^2 - (2\sqrt{6})^2} \Rightarrow \frac{5-2\sqrt{6}}{25-24} \Rightarrow \boxed{5-2\sqrt{6}}$

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Ex:- $x = 5\sqrt{3} + 4\sqrt{2}$

$$\frac{1}{x} = \frac{5\sqrt{3} - 4\sqrt{2}}{75 - 32}$$

$$= \frac{5\sqrt{3} - 4\sqrt{2}}{43}$$

Ex:- $x = 2\sqrt{10} - 6$

$$\frac{1}{x} = \frac{2\sqrt{10} + 6}{40 - 36} \Rightarrow \frac{2\sqrt{10} + 6}{4} = \frac{\sqrt{10} + 3}{2}$$

41 When $(4 + \sqrt{7})$ is presented in the form of perfect square it will be equal to:

- (a) $(2 + \sqrt{7})^2$ (b) $\left(\frac{\sqrt{7}}{2} + \frac{1}{2}\right)^2$ (c) $\left\{\frac{1}{\sqrt{2}}(\sqrt{7}+1)\right\}^2$ (d) $(\sqrt{3} + \sqrt{4})^2$

Soln:-

$$4 + \sqrt{7} \Rightarrow \frac{8 + 2\sqrt{7}}{2} \Rightarrow$$

$$\Rightarrow \frac{(1 + \sqrt{7})^2}{2}$$

$$\Rightarrow \left[\frac{1}{\sqrt{2}}(\sqrt{7}+1)\right]^2 \underline{\text{Ans}}$$

$$\begin{array}{c} 2\sqrt{7} \\ \swarrow \quad \searrow \\ \textcircled{2} \quad \sqrt{7} \\ \quad \swarrow \quad \searrow \\ \quad 1 \quad \sqrt{7} \\ \quad \quad \quad 1 + 7 = 8 \end{array}$$

42 $(\sqrt{2} + \sqrt{7 - 2\sqrt{10}})$ is equal to:

- (a) $\sqrt{2}$ (b) $\sqrt{7}$ (c) $\sqrt{5}$ (d) $2\sqrt{5}$

Soln:-

$$\sqrt{2} + \sqrt{(5 - \sqrt{2})^2}$$

$$\Rightarrow \sqrt{2} + \sqrt{5} - \sqrt{2}$$

$$\Rightarrow \sqrt{5} \underline{\text{Ans}}$$

$$\begin{array}{c} 2\sqrt{10} \\ \swarrow \quad \searrow \\ \textcircled{2} \quad \sqrt{10} \\ \quad \swarrow \quad \searrow \\ \quad \sqrt{5} \quad \sqrt{2} \end{array}$$

43 The value of $\sqrt{28 + 10\sqrt{3}} - \sqrt{7 - 4\sqrt{3}}$ is closest to:

- (a) 7.2 (b) 5.8 (c) 6.1 (d) 6.5

Soln:-

$$\sqrt{28 + 10\sqrt{3}} - \sqrt{7 - 4\sqrt{3}}$$

$$\Rightarrow (5 + \sqrt{3}) - (2 - \sqrt{3})$$

$$\Rightarrow 3 + 2\sqrt{3}$$

$$\Rightarrow 3 + 3.4$$

$$\approx \boxed{6.5} \underline{\text{Ans}}$$

[44] If $\sqrt{11-3\sqrt{8}} = a+b\sqrt{2}$, then what is the value of $(2a+3b)$?

Soln:-

$$\Rightarrow 3-\sqrt{2}$$

$$a=3$$

$$b=-1$$

$$(2a+3b) \Rightarrow 6-3$$

$$= \boxed{3} \text{ Ans}$$

$$\begin{array}{c} 3\sqrt{8} \\ \Downarrow \\ 3 \times 2\sqrt{2} \\ \swarrow \quad \searrow \\ \textcircled{2} \quad 3\sqrt{2} \\ \quad \swarrow \quad \searrow \\ \quad 3 \quad \sqrt{2} \end{array}$$

[45] The value of $\frac{1}{\sqrt{17+12\sqrt{2}}}$ is closet to:

(a) 1.4

(b) 1.2

(c) 0.14

(d) 0.17

Soln

$$\frac{1}{\sqrt{17+12\sqrt{2}}} \Rightarrow \frac{1}{3+2\sqrt{2}}$$

$$\begin{array}{c} 12\sqrt{2} \\ \swarrow \quad \searrow \\ \textcircled{2} \quad 6\sqrt{2} \\ \quad \swarrow \quad \searrow \\ \quad 3 \quad \textcircled{2}\sqrt{2} \\ \quad \quad 9+8=17 \end{array}$$

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[46] If $x = \sqrt{-\sqrt{3} + \sqrt{3+8\sqrt{7+4\sqrt{3}}}}$, where $x > 0$, then

the value x is equal to;

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Soln:-

$$\Rightarrow \sqrt{-\sqrt{3} + \sqrt{3+8(2+\sqrt{3})}}$$

$$\Rightarrow \sqrt{-\sqrt{3} + \sqrt{19+8\sqrt{3}}}$$

$$\Rightarrow \sqrt{-\cancel{\sqrt{3}} + 4 + \cancel{\sqrt{3}}}$$

$$\Rightarrow \sqrt{4}$$

$$\Rightarrow \boxed{2} \text{ Ans}$$

$$\begin{array}{c} 8\sqrt{3} \\ \swarrow \quad \searrow \\ 2 \times 4\sqrt{3} \\ \quad \swarrow \quad \searrow \\ \quad 4 \quad \sqrt{3} \end{array}$$

47] The value of

$$\frac{1}{\sqrt{12-\sqrt{40}}} - \frac{1}{\sqrt{8-\sqrt{60}}} - \frac{2}{\sqrt{10+\sqrt{84}}}$$

Soln:- (a) 0 (b) 1 (c) 2 (d) 3

$$\Rightarrow \frac{1}{(\sqrt{7}-\sqrt{5})} - \frac{1}{(\sqrt{5}-\sqrt{3})} - \frac{2}{(\sqrt{7}+\sqrt{3})}$$

$$\Rightarrow \frac{\sqrt{7}+\sqrt{5}}{2} - \frac{\sqrt{5}+\sqrt{3}}{2} - \frac{(\sqrt{7}-\sqrt{3})}{2}$$

$$\Rightarrow \frac{1}{2} (\sqrt{7}+\sqrt{5} - \sqrt{5}-\sqrt{3} - \sqrt{7}+\sqrt{3})$$

$$\Rightarrow \boxed{0} \text{ ans}$$

48] If $\frac{4+3\sqrt{3}}{\sqrt{7}+4\sqrt{3}} = A + \sqrt{B}$, then $B-A$ is

(a) -13 (b) $2\sqrt{3}$ (c) 13 (d) $3\sqrt{3} - \sqrt{7}$

Soln:- $\frac{4+3\sqrt{3}}{(2+\sqrt{3})} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}$

$$\Rightarrow \frac{(4+3\sqrt{3})(2-\sqrt{3})}{4-3}$$

$$\Rightarrow 8-4\sqrt{3}+6\sqrt{3}-9$$

$$\Rightarrow 2\sqrt{3}-1 \Rightarrow A + \sqrt{B}$$

$$A = -1$$

$$\sqrt{B} = 2\sqrt{3}$$

$$B = 12$$

$$B-A = 12+1 = \boxed{13} \text{ ans}$$

49 $\frac{6^2 + 7^2 + 8^2 + 9^2 + 10^2}{\sqrt{7+4\sqrt{3}} - \sqrt{4+2\sqrt{3}}} = ?$

- (a) 330 (b) 305 (c) 355 (d) 306

Soln:-

$$\frac{36 + 49 + 64 + 81 + 100}{(2 + \sqrt{3}) - (\sqrt{3} + 1)}$$

$$\Rightarrow \frac{330}{1} = \boxed{330} \text{ Ans}$$

50 $\frac{\sqrt{26 - 15\sqrt{3}}}{5\sqrt{2} - \sqrt{38 + 5\sqrt{3}}} = ?$

- (a) $\sqrt{2}$ (b) $\frac{1}{\sqrt{3}}$ (c) $\sqrt{3}$ (d) $\frac{1}{\sqrt{2}}$

Soln:- multiply and divide by $\sqrt{2}$

$$\frac{\sqrt{52 - 2 \times 15\sqrt{3}}}{10 - \sqrt{76 + 2 \times 5\sqrt{3}}}$$

$$\Rightarrow \frac{3\sqrt{3} - 5}{10 - (5\sqrt{3} + 1)}$$

$$\Rightarrow \frac{(3\sqrt{3} - 5)}{\sqrt{3}(3\sqrt{3} - 5)}$$

$$\Rightarrow \boxed{\frac{1}{\sqrt{3}}} \text{ Ans}$$

$$\Rightarrow \begin{aligned} (a+b+c)^2 &= a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \\ (a-b+c)^2 &= a^2 + b^2 + c^2 - 2ab - 2bc + 2ca \\ (a+b-c)^2 &= a^2 + b^2 + c^2 + 2ab - 2bc - 2ca \end{aligned}$$

[51] If $\sqrt{15} + \sqrt{60} + \sqrt{84} + \sqrt{140} = \sqrt{a} + \sqrt{b} + \sqrt{c}$, then the value of $a+b+c$?

Soln:-

$$\begin{array}{cccc} 15 & + & \sqrt{60} & + & \sqrt{84} & + & \sqrt{140} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ a+b+c & & 2 \times \sqrt{5} \times \sqrt{3} & & 2 \times \sqrt{3} \times \sqrt{7} & & 2 \times \sqrt{7} \times \sqrt{5} \end{array}$$

$$\therefore \sqrt{a} + \sqrt{b} + \sqrt{c} = \sqrt{5} + \sqrt{3} + \sqrt{7}$$

$$a+b+c = \boxed{15} \quad \underline{\text{Ans}}$$

[52] The expression $\sqrt{10 + 2(\sqrt{6} - \sqrt{15} - \sqrt{10})}$ is equal to:

(a) $\sqrt{3} - \sqrt{2} - \sqrt{5}$ (b) $\sqrt{3} - \sqrt{2} + \sqrt{5}$ (c) $\sqrt{2} - \sqrt{3} - \sqrt{5}$ (d) $\sqrt{3} + \sqrt{2} - \sqrt{5}$

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Soln:-

$$\begin{array}{cccc} \sqrt{10 + 2(\sqrt{6} - \sqrt{15} - \sqrt{10})} \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ a^2+b^2+c^2 \quad \sqrt{2} \times \sqrt{3} \quad \sqrt{3} \times \sqrt{5} \quad \sqrt{5} \times \sqrt{2} \end{array}$$

$$\Rightarrow \sqrt{(\sqrt{3} + \sqrt{2} - \sqrt{5})^2}$$

$$\Rightarrow \boxed{\sqrt{3} + \sqrt{2} - \sqrt{5}} \quad \underline{\text{Ans}}$$

[53] The expression $\frac{15(\sqrt{10} + \sqrt{5})}{\sqrt{10} + \sqrt{20} + \sqrt{40} - \sqrt{5} - \sqrt{80}}$ is equal to:

- (a) $10(3+2\sqrt{5})$ (b) $5+2\sqrt{2}$ (c) $5(3+2\sqrt{2})$ (d) $5-2\sqrt{5}$

Soln:-

$$\begin{aligned} & \frac{15(\sqrt{10} + \sqrt{5})}{\sqrt{10} + 2\sqrt{5} + 2\sqrt{10} - \sqrt{5} - 4\sqrt{5}} \\ \Rightarrow & \frac{15(\sqrt{10} + \sqrt{5})}{3(\sqrt{10} - \sqrt{5})} \\ \Rightarrow & \frac{\cancel{3}(\sqrt{10} + \sqrt{5})(\sqrt{10} + \sqrt{5})}{\cancel{3}} \Rightarrow \frac{(\sqrt{10} + \sqrt{5})^2}{\Rightarrow \boxed{5(3+2\sqrt{2})}} \text{ Ans} \end{aligned}$$

[54] If $\frac{1}{4-\sqrt{8}} + \frac{3+2\sqrt{2}}{3-2\sqrt{2}} - \frac{3-2\sqrt{2}}{3+2\sqrt{2}} = a + b\sqrt{2}$ then what is the value of $(3a+4b)$?

Soln:- $\frac{4+\sqrt{8}}{8} + \frac{(3+2\sqrt{2})^2 - (3-2\sqrt{2})^2}{1} = a + b\sqrt{2}$

$$\Rightarrow \frac{1}{2} + \frac{\sqrt{2}}{4} + 4 \times 3 \times 2\sqrt{2} \quad \left[\because (a+b)^2 - (a-b)^2 = 4ab \right]$$

$$\Rightarrow \frac{1}{2} + \frac{97\sqrt{2}}{4} = a + b\sqrt{2}$$

$$a = \frac{1}{2}, \quad b = \frac{97}{4}$$

$$(3a+4b) \Rightarrow 3 \times \frac{1}{2} + 4 \times \frac{97}{4}$$

$$\Rightarrow \frac{3}{2} + 97$$

$$\Rightarrow \boxed{98\frac{1}{2}} \text{ Ans}$$

55 The value of

$$5 - \frac{8+2\sqrt{5}}{4} - \frac{1}{8+2\sqrt{5}} \text{ is equal to:}$$

(a) $\frac{1}{4}$

(b) 1

(c) $\frac{2}{3}$

(d) $\frac{1}{2}$

Soln:-

$$5 - \left[\frac{8+2\sqrt{5}}{4} + \frac{8-2\sqrt{5}}{4} \right]$$

$$\Rightarrow 5 - \frac{1}{4}(16) \Rightarrow 5 - 4 \Rightarrow \boxed{1} \text{ Ans}$$

56 The value of

$$\frac{1}{(9+4\sqrt{5})^2} + \frac{1}{(9-4\sqrt{5})^2} \text{ is:}$$

Soln:-

$$\frac{(9-4\sqrt{5})^2 + (9+4\sqrt{5})^2}{1} \quad [(a+b)^2 + (a-b)^2 = 2(a^2+b^2)]$$

$$\Rightarrow 2(81+80) = \boxed{322} \text{ Ans}$$

57 The value of $\frac{1}{(3+2\sqrt{2})^3} + \frac{1}{(3-2\sqrt{2})^3}$ is

Soln:-

$$(3-2\sqrt{2})^3 + (3+2\sqrt{2})^3$$

$\Rightarrow$

$$6 \left[\underbrace{17}_{a^2} - \underbrace{12\sqrt{2}}_{ab} + \underbrace{17}_{b^2} + \underbrace{12\sqrt{2}}_{ab} - \underbrace{1}_{ab} \right]$$

$$[\because a^3+b^3 = (a+b)(a^2+b^2-ab)]$$

$$\Rightarrow 6[17+17-1]$$

$$\Rightarrow 6 \times 33$$

$$\Rightarrow \boxed{198} \text{ Ans}$$

$$\boxed{58} \quad \left(\frac{2+\sqrt{3}}{2-\sqrt{3}} + \frac{2-\sqrt{3}}{2+\sqrt{3}} + \frac{\sqrt{3}-1}{\sqrt{3}+1} \right)$$

Soln:-

$$\Rightarrow \frac{(2+\sqrt{3})^2 + (2-\sqrt{3})^2}{1} + \frac{(\sqrt{3}-1)^2}{2} \quad [\because (a+b)^2 + (a-b)^2 = 2(a^2+b^2)]$$

$$\Rightarrow 2(7) + \frac{4-2\sqrt{3}}{2}$$

$$\Rightarrow 14 + 2 - \sqrt{3} \Rightarrow \boxed{16 - \sqrt{3}} \text{ Ans}$$

$$\boxed{59} \quad \left(\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} \right)^2 + \left(\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}} \right)^2 \text{ is equal to:}$$

Soln:-

$$\frac{8+2\sqrt{15}}{8-2\sqrt{15}} + \frac{8-2\sqrt{15}}{8+2\sqrt{15}}$$

$$\Rightarrow \frac{4+\sqrt{15}}{4-\sqrt{15}} + \frac{4-\sqrt{15}}{4+\sqrt{15}}$$

$$\Rightarrow \frac{(4+\sqrt{15})^2 + (4-\sqrt{15})^2}{(16-15)} \Rightarrow 2(16+15) \Rightarrow \boxed{62} \text{ Ans}$$

$$\boxed{60} \quad \left(\frac{1+\sqrt{2}}{\sqrt{5}+\sqrt{3}} + \frac{1-\sqrt{2}}{\sqrt{5}-\sqrt{3}} \right) \text{ simplifies to:}$$

(a) $\sqrt{5} + \sqrt{6}$ (b) $2\sqrt{5} + \sqrt{6}$ (c) $\sqrt{5} - \sqrt{6}$ (d) $2\sqrt{5} - 3\sqrt{6}$

Soln:-

$$\frac{(1+\sqrt{2})(\sqrt{5}-\sqrt{3}) + (1-\sqrt{2})(\sqrt{5}+\sqrt{3})}{(5-3)}$$

$$\Rightarrow \frac{1}{2} [\cancel{\sqrt{5}} - \cancel{\sqrt{3}} + \sqrt{10} - \sqrt{6} + \sqrt{5} + \cancel{\sqrt{3}} - \sqrt{10} - \sqrt{6}]$$

$$\Rightarrow \frac{1}{2} [2\sqrt{5} - 2\sqrt{6}]$$

$$\Rightarrow \boxed{\sqrt{5} - \sqrt{6}} \text{ Ans}$$

[61] The value of

$$\frac{3\sqrt{7}}{\sqrt{5} + \sqrt{2}} - \frac{5\sqrt{5}}{\sqrt{2} + \sqrt{7}} + \frac{2\sqrt{2}}{\sqrt{7} + \sqrt{5}} \text{ is :}$$

- (a) 1 (b) 0 (c) $2\sqrt{3}$ (d) $\sqrt{7}$

Solⁿ:-
$$\frac{3\sqrt{7}(\sqrt{5}-\sqrt{2})}{3} - \frac{5\sqrt{5}(\sqrt{7}-\sqrt{2})}{5} + \frac{2\sqrt{2}(\sqrt{7}-\sqrt{5})}{2}$$

$$\cancel{\sqrt{35}} - \cancel{\sqrt{14}} - \cancel{\sqrt{35}} + \cancel{\sqrt{10}} + \cancel{\sqrt{14}} - \cancel{\sqrt{10}} = \boxed{0} \text{ Ans}$$

[62] If x, y are rational numbers and

$$\frac{5 + \sqrt{11}}{3 - 2\sqrt{11}} = x + y\sqrt{11}. \text{ The values of } x \text{ and } y \text{ are.}$$

Solⁿ:-
$$\frac{(5 + \sqrt{11})(3 + 2\sqrt{11})}{(9 - 44)} = x + y\sqrt{11}$$

$$\Rightarrow \frac{15 + 10\sqrt{11} + 3\sqrt{11} + 22}{-35}$$

$$\Rightarrow \frac{37 + 13\sqrt{11}}{-35} = x + y\sqrt{11}$$

$$\Rightarrow -\frac{37}{35} + -\frac{13}{35}\sqrt{11} \text{ Compare } x = -\frac{37}{35}, y = -\frac{13}{35} \text{ Ans}$$

[63] The simplified value of

$$(\sqrt{3}+1)(10+\sqrt{12})(\sqrt{12}-2)(5-\sqrt{3}) \text{ is:}$$

Solⁿ:-
$$(10\sqrt{3} + \sqrt{36} + 10 + \sqrt{12})(5\sqrt{12} - \sqrt{36} - 10 + 2\sqrt{3})$$

$$\Rightarrow (10\sqrt{3} + 6 + 10 + 2\sqrt{3})(10\sqrt{3} - 6 - 10 + 2\sqrt{3})$$

$$\Rightarrow (12\sqrt{3} + 16)(12\sqrt{3} - 16)$$

$$\Rightarrow 144 \times 3 - 256$$

$$\Rightarrow \boxed{176} \text{ Ans}$$

64 The simplified value of

$$(\sqrt{6} + \sqrt{10} - \sqrt{21} - \sqrt{35})(\sqrt{6} - \sqrt{10} + \sqrt{21} - \sqrt{35}) \text{ is:}$$

(a) 13

(b) 12

(c) 11

(d) 10

Soln:- $\{(\sqrt{6} - \sqrt{35}) + (\sqrt{10} - \sqrt{21})\} \{(\sqrt{6} - \sqrt{35}) - (\sqrt{10} - \sqrt{21})\}$

$$\Rightarrow (\sqrt{6} - \sqrt{35})^2 - (\sqrt{10} - \sqrt{21})^2$$

$$\Rightarrow (6 + 35 - 2\sqrt{210}) - (10 + 21 - 2\sqrt{210})$$

$$\Rightarrow 41 - 31$$

$$\Rightarrow \boxed{10} \text{ Ans}$$

65 Let $x = \sqrt[3]{27} - \sqrt{6\frac{3}{4}}$ and $y = \frac{\sqrt{45} + \sqrt{605} + \sqrt{245}}{\sqrt{80} + \sqrt{125}}$,

then the value of $x^2 + y^2$ is:

Soln:-

$$x = (27)^{\frac{1}{3}} - \sqrt{6\frac{3}{4}}$$

$$\Rightarrow ((3)^3)^{\frac{1}{3}} - \frac{3\sqrt{3}}{2}$$

$$\Rightarrow \sqrt{3} - \frac{3\sqrt{3}}{2}$$

$$\Rightarrow \frac{-\sqrt{3}}{2}$$

$$y = \frac{3\sqrt{5} + 11\sqrt{5} + 7\sqrt{5}}{4\sqrt{5} + 5\sqrt{5}}$$

$$= \frac{7}{3}$$

$$x^2 + y^2 = \frac{3}{4} + \frac{49}{9}$$

$$= \boxed{\frac{223}{36}} \text{ Ans}$$

[66] The value of

$$\frac{1}{\sqrt{7}-\sqrt{6}} - \frac{1}{\sqrt{6}-\sqrt{5}} + \frac{1}{\sqrt{5}-2} - \frac{1}{\sqrt{8}-\sqrt{7}} + \frac{1}{3-\sqrt{8}} \text{ is:}$$

Solⁿ $\sqrt{7}+\sqrt{6} - (\sqrt{6}+\sqrt{5}) + (\sqrt{5}+2) - (\sqrt{8}+\sqrt{7}) + (3+\sqrt{8})$

$\Rightarrow$ [5] Ans

[67] If $\frac{4}{1+\sqrt{2}+\sqrt{3}} = a+b\sqrt{2}+c\sqrt{3}-d\sqrt{6}$, where a, b, c, d

are natural numbers, then the value of $a+b+c+d$.

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Solⁿ:- $\frac{4}{[(\sqrt{3}+\sqrt{2})+1]} \times \frac{(\sqrt{3}+\sqrt{2})-1}{[(\sqrt{3}+\sqrt{2})-1]}$

$\Rightarrow \frac{4[(\sqrt{3}+\sqrt{2})-1]}{(\sqrt{3}+\sqrt{2})^2-1}$

$\Rightarrow \frac{4[\sqrt{3}+\sqrt{2}-1]}{4+2\sqrt{6}}$

$\Rightarrow \frac{[2\sqrt{3}+2\sqrt{2}-2]}{2+\sqrt{6}}$

$\Rightarrow \frac{(2\sqrt{3}+2\sqrt{2}-2)(\sqrt{6}-2)}{6-4}$

$\Rightarrow \frac{6\sqrt{2}+4\sqrt{3}-2\sqrt{6}-4\sqrt{3}-4\sqrt{2}+4}{2}$

$\Rightarrow \frac{2\sqrt{2}-2\sqrt{6}+4}{2}$

$\Rightarrow \sqrt{2}-\sqrt{6}+2 = a+b\sqrt{2}+c\sqrt{3}-d\sqrt{6}$

Compare $a=2, b=1, c=0, d=1$

$a+b+c+d = 2+1+0+1 =$ [4] Ans

[68] The value of $5\sqrt{3} + 7\sqrt{2} - \sqrt{6} - \frac{23}{\sqrt{2} + \sqrt{3} + \sqrt{6}}$ is:

Solⁿ:- [m] Basic

$$5\sqrt{3} + 7\sqrt{2} - \sqrt{6} - 23 \left[\frac{(\sqrt{2} + \sqrt{3}) - \sqrt{6}}{(\sqrt{2} + \sqrt{3})^2 - (\sqrt{6})^2} \right]$$

$$\Rightarrow 5\sqrt{3} + 7\sqrt{2} - \sqrt{6} - 23 \left[\frac{(\sqrt{2} + \sqrt{3} - \sqrt{6})}{2\sqrt{6} - 1} \right]$$

$$\Rightarrow 5\sqrt{3} + 7\sqrt{2} - \sqrt{6} - 23 \left[\frac{(\sqrt{2} + \sqrt{3} - \sqrt{6})(2\sqrt{6} + 1)}{(2\sqrt{6})^2 - 1} \right]$$

$$\Rightarrow 5\sqrt{3} + 7\sqrt{2} - \sqrt{6} - 23 \left[\frac{4\sqrt{3} + \sqrt{2} + 6\sqrt{2} + \sqrt{3} - 12 - \sqrt{6}}{23} \right]$$

$$\Rightarrow \boxed{12} \text{ Ans}$$

[69] $\frac{12}{3 + \sqrt{5} + 2\sqrt{2}}$ is equal to

(a) $1 - \sqrt{5} + \sqrt{2} + \sqrt{10}$

(b) $1 + \sqrt{5} + \sqrt{2} - \sqrt{10}$

(c) $1 + \sqrt{5} + \sqrt{2} + \sqrt{10}$

(d) $1 - \sqrt{5} - \sqrt{2} + \sqrt{10}$

Solⁿ:- $\frac{12[(3 + \sqrt{5}) - 2\sqrt{2}]}{(3 + \sqrt{5})^2 - (2\sqrt{2})^2}$

$$\Rightarrow \frac{12(3 + \sqrt{5} - 2\sqrt{2})}{9 + 5 + 6\sqrt{5} - 8}$$

$$\Rightarrow \frac{12(3 + \sqrt{5} - 2\sqrt{2})}{6\sqrt{5} + 6} \Rightarrow \frac{2(3 + \sqrt{5} - 2\sqrt{2})}{\sqrt{5} + 1}$$

$$\Rightarrow \frac{2(3 + \sqrt{5} - 2\sqrt{2})(\sqrt{5} - 1)}{4}$$

$$\Rightarrow \frac{2(3\sqrt{5} - 3 + 5 - \sqrt{5} - 2\sqrt{10} + 2\sqrt{2})}{4}$$

$$\Rightarrow \frac{2 \times 2(\sqrt{5} + \sqrt{2} - \sqrt{10} + 1)}{4}$$

$\boxed{\sqrt{5} + \sqrt{2} - \sqrt{10} + 1}$ Ans

70 $\left[\frac{1}{\sqrt{2} + \sqrt{3} - \sqrt{5}} + \frac{1}{\sqrt{2} - \sqrt{3} - \sqrt{5}} \right]$ in simplified form equals to:

Soln:- $\frac{1}{\sqrt{2} + \sqrt{3} - \sqrt{5}} \Rightarrow \frac{(\sqrt{2} + \sqrt{3}) + \sqrt{5}}{(\sqrt{2} + \sqrt{3})^2 - (\sqrt{5})^2} \Rightarrow \frac{\sqrt{2} + \sqrt{3} + \sqrt{5}}{2\sqrt{6}}$

$\frac{1}{\sqrt{2} - \sqrt{3} - \sqrt{5}} \Rightarrow \frac{(\sqrt{2} - \sqrt{3}) + \sqrt{5}}{(\sqrt{2} - \sqrt{3})^2 - (\sqrt{5})^2} \Rightarrow \frac{\sqrt{2} - \sqrt{3} + \sqrt{5}}{-2\sqrt{6}}$

So $\frac{(\sqrt{2} + \sqrt{3} + \sqrt{5})}{2\sqrt{6}} - \frac{(\sqrt{2} - \sqrt{3} + \sqrt{5})}{2\sqrt{6}}$

$\Rightarrow \frac{\cancel{\sqrt{2}} + \sqrt{3} + \cancel{\sqrt{5}} - \cancel{\sqrt{2}} + \sqrt{3} - \cancel{\sqrt{5}}}{2\sqrt{6}} \Rightarrow \frac{2\sqrt{3}}{2\sqrt{6}} = \boxed{\frac{1}{\sqrt{2}}}$ Ans

71 If $x = 1 + \sqrt{2} + \sqrt{3}$, then find the value of $x^2 - 2x + 4$.

Soln:- $(x-1)^2 = (\sqrt{2} + \sqrt{3})^2$
 $x^2 - 2x + 1 = 2 + 3 + 2\sqrt{6}$

$x^2 - 2x + 4 = 8 + 2\sqrt{6} \Rightarrow \boxed{2(4 + \sqrt{6})}$ Ans

72 If $(\sqrt{2} + \sqrt{5} - \sqrt{3}) \times k = -12$, then what will be the value of k ?

(a) $(\sqrt{2} + \sqrt{5} - \sqrt{3})(2 + \sqrt{5})$

(b) $(\sqrt{2} + \sqrt{5} + \sqrt{3})(2 - \sqrt{5})$

(c) $(\sqrt{2} + \sqrt{5} + \sqrt{3})(2 - \sqrt{10})$

(d) $(\sqrt{2} + \sqrt{5} + \sqrt{3})$

Soln:- $k = \frac{-12}{(\sqrt{2} + \sqrt{5} - \sqrt{3})} \Rightarrow \frac{-12[\sqrt{2} + \sqrt{5} + \sqrt{3}]}{(\sqrt{2} + \sqrt{5})^2 - (\sqrt{3})^2} \Rightarrow \frac{-12[\sqrt{2} + \sqrt{5} + \sqrt{3}]}{2(2 + \sqrt{10})}$

$\Rightarrow \frac{-6[\sqrt{2} + \sqrt{5} + \sqrt{3}](\sqrt{10} - 2)}{(\sqrt{10})^2 - 2^2} \Rightarrow \boxed{(\sqrt{2} + \sqrt{5} + \sqrt{3})(2 - \sqrt{10})}$ Ans

73] If $a = \frac{\sqrt{3}}{2}$, then the value of $\sqrt{1+a} + \sqrt{1-a}$ is:

Soln:- let $x = \sqrt{1+\frac{\sqrt{3}}{2}} + \sqrt{1-\frac{\sqrt{3}}{2}}$

$$x^2 = 2 - 2\sqrt{1-\frac{3}{4}}$$

$$x^2 = 2 - 2 \times \frac{1}{2}$$

$$x^2 = 1$$

$$\boxed{x = 1} \text{ Ans}$$

74] If $a = \frac{\sqrt{5}+1}{\sqrt{5}-1}$, $b = \frac{\sqrt{5}-1}{\sqrt{5}+1}$, then the value of $\left(\frac{a^2+ab+b^2}{a^2-ab+b^2}\right)$ is:

Soln:- $a+b = \frac{\sqrt{5}+1}{\sqrt{5}-1} + \frac{\sqrt{5}-1}{\sqrt{5}+1}$

$$= \frac{(\sqrt{5}+1)^2 + (\sqrt{5}-1)^2}{(\sqrt{5})^2 - (1)^2} \Rightarrow \frac{2[(\sqrt{5})^2 + 1]}{5-1} = \frac{2[5+1]}{4}$$

$$= \boxed{3}$$

$$ab = \frac{\sqrt{5}+1}{\sqrt{5}-1} \times \frac{\sqrt{5}-1}{\sqrt{5}+1} = \boxed{1}$$

$$\frac{a^2+ab+b^2}{a^2-ab+b^2} \Rightarrow \frac{(a+b)^2 - ab}{(a+b)^2 - 3ab}$$

$$\Rightarrow \frac{3^2 - 1}{3^2 - 3}$$

$$\Rightarrow \frac{9-1}{9-3}$$

$$\Rightarrow \boxed{\frac{4}{3}} \text{ Ans}$$

75] If $\frac{x + \sqrt{x^2 - 1}}{x - \sqrt{x^2 - 1}} + \frac{x - \sqrt{x^2 - 1}}{x + \sqrt{x^2 - 1}} = 194$, then $x = ?$

Soln:-

$$\frac{(x + \sqrt{x^2 - 1})^2 + (x - \sqrt{x^2 - 1})^2}{x^2 - (x^2 - 1)} = 194$$

$$(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$$

$$\Rightarrow \frac{2(x^2 + x^2 - 1)}{1} = 194$$

$$\Rightarrow 4x^2 = 194$$

$$x^2 = 49$$

$$\boxed{x = 7} \text{ Ans}$$

76] If $\sqrt{5x-6} + \sqrt{5x+6} = 6$, then $x = ?$

(a) -4

(b) 0

(c) 2

(d) 4

Soln:- m-I Basic:-

Square both side

$$\Rightarrow 5x-6 + 5x+6 + 2\sqrt{25x^2-36} = 36$$

$$\Rightarrow 10x + 2\sqrt{25x^2-36} = 36$$

$$\Rightarrow \sqrt{25x^2-36} = (8-5x)$$

Both square

$$25x^2 - 36 = 324 + 25x^2 - 180x$$

$$180x = 324 + 36$$

$$x = \frac{360}{180}$$

$$\boxed{x = 2} \text{ Ans}$$

m-II Option method:-

Put $x = 2$

$$\sqrt{10-6} + \sqrt{10+6}$$

$$2 + 4 = \boxed{6}$$

satisfy ✓

Componendo & Dividendo :-

$$\frac{a}{b} = \frac{c}{d}$$

apply C & D (add and subtracted denominators in num.)

$$\boxed{\frac{a+b}{a-b} = \frac{c+d}{c-d}}$$

again apply

$$\frac{(a+b)+(a-b)}{(a+b)-(a-b)} = \frac{(c+d)+(c-d)}{(c+d)-(c-d)}$$

$$\frac{2a}{2b} = \frac{2c}{2d}$$

$$\boxed{\frac{a}{b} = \frac{c}{d}}$$

$\Rightarrow$ If we apply C & D two times on a fraction, same is achieved.

So

$$\text{If } \left[\frac{a}{b} = \frac{c}{d} \Leftrightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d} \right]$$

$$\Rightarrow \left[\frac{a}{b} = \frac{c}{d} \Leftrightarrow \frac{a-b}{a+b} = \frac{c-d}{c+d} \right]$$

$$\Rightarrow \left[\frac{a}{b} = -\frac{c}{d} \Leftrightarrow \frac{a+b}{a-b} = \frac{c-d}{c+d} \right]$$

$$\Rightarrow \left[-\frac{a}{b} = \frac{c}{d} \Leftrightarrow \frac{a-b}{a+b} = \frac{c+d}{c-d} \right]$$

77 If $\frac{\sqrt{5+x} + \sqrt{5-x}}{\sqrt{5+x} - \sqrt{5-x}} = \frac{3}{1}$, then what is the value of x ?

Solⁿ:- Apply C & D

$$\frac{\sqrt{5+x}}{\sqrt{5-x}} = \frac{3+1}{3-1}$$

$$\frac{\sqrt{5+x}}{\sqrt{5-x}} = \frac{4}{2} \Rightarrow 2$$

Square

$$\frac{5+x}{5-x} = 4$$

again C & D $\Rightarrow \frac{5}{x} = \frac{5}{3} \Rightarrow \boxed{x=3}$ Ans

78 If $p = \frac{\sqrt{3q+2} + \sqrt{3q-2}}{\sqrt{3q+2} - \sqrt{3q-2}}$, then the value of $p^2 - 3pq + 2$ is:

Solⁿ:- C & D

$$\frac{p+1}{p-1} = \frac{\sqrt{3q+2}}{\sqrt{3q-2}}$$

Square

$$\frac{(p+1)^2}{(p-1)^2} = \frac{3q+2}{3q-2}$$

again C & D

$$\frac{(p+1)^2 + (p-1)^2}{(p+1)^2 - (p-1)^2} = \frac{3q}{2}$$

$$\Rightarrow \frac{2(p^2+1)}{4p} = \frac{3q}{2}$$

$$\Rightarrow p^2 + 1 = 3pq$$

$$\Rightarrow p^2 - 3pq + 1 = 0 \Rightarrow p^2 - 3pq + 2 = \boxed{1} \text{ Ans}$$

[79] Given that $\sqrt{3} = 1.732$, Then the value of

$$\frac{3 + \sqrt{6}}{5\sqrt{3} - 2\sqrt{12} - \sqrt{32} + \sqrt{50}} \text{ is:}$$

- (a) 4.899 (b) 2.551 (c) 1.414 (d) 1.732

Soln:-

$$\frac{3 + \sqrt{6}}{5\sqrt{3} - 4\sqrt{3} - 4\sqrt{2} + 5\sqrt{2}}$$

$$\Rightarrow \frac{3 + \sqrt{6}}{\sqrt{3} + \sqrt{2}} \Rightarrow \frac{\sqrt{3}(\sqrt{3} + \sqrt{2})}{(\sqrt{3} + \sqrt{2})} \Rightarrow \sqrt{3} = \boxed{1.732} \text{ Ans}$$

[80] If $\sqrt{2} = 1.4142$, find the value of

$$2\sqrt{2} + \sqrt{2} + \frac{1}{2 + \sqrt{2}} - \frac{1}{2 - \sqrt{2}}$$

- (a) 1.4144 (b) 2.8284 (c) 28.284 (d) 2.4142

Soln:-

$$2\sqrt{2} + \sqrt{2} + \left(\frac{2 - \sqrt{2} - 2 - \sqrt{2}}{4 - 2} \right)$$

$$2\sqrt{2} + \sqrt{2} - \frac{2\sqrt{2}}{2} \Rightarrow 3\sqrt{2} - \sqrt{2} = 2\sqrt{2} = 2 \times 1.4142 = \boxed{2.8284} \text{ Ans}$$

[81] If $\sqrt{33} = 5.745$, then the value of $\sqrt{\frac{3}{11}}$ is:

- (a) 0.5223 (b) 6.32 (c) 2.035 (d) 1

Soln

$$\sqrt{\frac{3}{11}} \Rightarrow \sqrt{\frac{3}{11} \times \frac{11}{11}}$$

$$\Rightarrow \frac{\sqrt{33}}{11}$$

$$\Rightarrow \frac{5.745}{11} = \boxed{0.5223} \text{ Ans}$$

[81] Find the value of x , if $21^{\sqrt{x}} + 20^{\sqrt{x}} = 29^{\sqrt{x}}$

Solⁿ:- Triplet :- 20, 21, 29

[CHSL - 2020]

$$20^2 + 21^2 = 29^2$$

Compare $\sqrt{x} = 2$

$$\boxed{x = 4} \text{ Ans}$$

[82] Find the value of x , if $3^{\sqrt[3]{x}} + 4^{\sqrt[3]{x}} = 5^{\sqrt[3]{x}}$

Solⁿ:- $3^2 + 4^2 = 5^2$

$$\sqrt[3]{x} = 2$$

$$x = 2^3 = \boxed{8} \text{ Ans}$$

[83] $3^x - 3^{x-1} = 486$, find x

Solⁿ:- $3^{x-1}(3-1) = 486$

$$3^{x-1} = 243$$

$$3^{x-1} = 3^5 \Rightarrow x-1=5 \Rightarrow \boxed{x=6} \text{ Ans}$$

[84] If $5^{x+1} - 5^{x-1} = 600$, then what is the value of 10^{2x} ?

Solⁿ:- $5^x(5 - 5^{-1}) = 600$

$$5^x \left(5 - \frac{1}{5}\right) = 600$$

$$5^x = \frac{600}{24} \times 5$$

$$5^x = 125 = 5^3$$

$$\boxed{x=3} \therefore 10^{2x} \Rightarrow \boxed{10^6} \text{ Ans}$$

85 If $25^{x-1} = 5^{2x-1} - 100$, then what is the value of x ?

Soln:- $5^{2x-2} = 5^{2x-1} - 100$
 $5^{2x-1} - 5^{2x-1} \times 5^{-1} = 100$

$$5^{2x-1} \left[1 - \frac{1}{5} \right] = 100$$

$$5^{2x-1} \times \frac{4}{5} = 100$$

$$5^{2x-1} = 5^3 \Rightarrow 2x-1 = 3$$

$$\Rightarrow \boxed{x=2} \text{ Ans}$$

86 If $9^{2x-1} - 81^{x-1} = 1944$, then x is

(a) 3

(b) $\frac{9}{4}$

(c) $\frac{4}{9}$

(d) $\frac{1}{3}$

Soln:- option method:-

(b) $9^{2 \times \frac{9}{4} - 1} - 81^{\frac{9}{4} - 1}$

$$\Rightarrow 9^{\frac{7}{2}} - 81^{\frac{5}{4}} \Rightarrow 3^7 - 3^5$$

Unit digit $\rightarrow 7 - 3 = \boxed{4} \checkmark$

87 If $9^x 3^y = 2187$ and $2^{3x} 2^{2y} - 4^{xy} = 0$, then $x+y = ?$

Soln:- $3^{2x+y} = 3^7$

$$2x+y = 7 \quad \text{--- (1)}$$

$$2^{3x+2y} = 2^{2xy}$$

$$3x+2y = 2xy \quad \text{--- (2)}$$

from (1) and (2)

Solve:- $2 \times \textcircled{2} + \textcircled{3} \Rightarrow 7$
 $\uparrow \quad \uparrow$

$$3 \times 2 + 2 \times 3 = 2 \times 2 \times 3$$

$$6 + 6 = 12$$

satisfy

$$x=2, y=3$$

$$\boxed{x+y = 5}$$

Ans

88 If $3^x = 4^y = 12^z$, then z is equal to:

(a) xy

(b) $x+y$

(c) $\frac{xy}{x+y}$

(d) $4x+3y$

Soln:- m-I Basic:-

$3^x = 4^y = 12^z = k$ (let)

$3^x = k \Rightarrow 3 = k^{\frac{1}{x}}$

$4^y = k \Rightarrow 4 = k^{\frac{1}{y}}$

$12^z = k \Rightarrow 12 = k^{\frac{1}{z}}$

$3 \times 4 = 12$

$k^{\frac{1}{x}} \cdot k^{\frac{1}{y}} = k^{\frac{1}{z}}$

$k^{\frac{1}{x} + \frac{1}{y}} = k^{\frac{1}{z}}$

$\Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$

m-II

$3^{\textcircled{1}} \times 4^{\textcircled{1}} = 12^{\textcircled{1}}$

Trick:-

$\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$

$z = \frac{xy}{x+y}$ Ans

89 If $2^x = 3^y = 18^z$, then

(a) $\frac{1}{x} + \frac{2}{y} - \frac{1}{z} = 0$

(b) $\frac{2}{x} + \frac{1}{y} + \frac{1}{z} = 0$

(c) $\frac{1}{x} + \frac{2}{y} + \frac{1}{z} = 0$

(d) $\frac{1}{x} - \frac{2}{y} + \frac{1}{z} = 0$

Soln:-

Trick:-

$2^{\textcircled{1}} \times 3^{\textcircled{2}} = 18^{\textcircled{1}}$

$\frac{1}{x} + \frac{2}{y} = \frac{1}{z} \Rightarrow$

$\frac{1}{x} + \frac{2}{y} + \frac{1}{z} = 0$

90 If $2^x = 3^y = 6^z$ then $\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$ is equal to:

Soln:- Trick:- $2^{\textcircled{1}} \times 3^{\textcircled{1}} = 6^{\textcircled{1}}$

$\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$

$\Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \boxed{0}$ Ans

91] If $x^{\frac{1}{6}} = y^{\frac{2}{3}}$, then relation between x and y is:
 (a) $x = y^2$ (b) $x = y^4$ (c) $x^4 = y$ (d) $x^3 = y^3$

Soln:-

$$x^{\frac{1}{6}} = y^{\frac{2}{3}}$$

$$x^3 = y^{2 \times 6}$$

$$\boxed{x = y^4} \text{ Ans}$$

92] If $f(x) = \frac{1}{x} - \frac{1}{x+1}$, then what is the value of
 $f(1) + f(2) + f(3) + \dots + f(10)$?

Soln:-

$$f(1) = \frac{1}{1} - \frac{1}{2}$$

$$f(2) = \frac{1}{2} - \frac{1}{3}$$

$$f(3) = \frac{1}{3} - \frac{1}{4}$$

$$\vdots$$

$$f(10) = \frac{1}{10} - \frac{1}{11}$$

$$\oplus \quad \frac{f(1) + f(2) + f(3) + \dots + f(10)}{\Rightarrow 1 - \frac{1}{11}} \Rightarrow \boxed{\frac{10}{11}} \text{ Ans}$$

93] If $(3x-2y) : (2x+3y) = 5 : 6$, then the value of
 $\left(\frac{3\sqrt{x} + 3\sqrt{y}}{3\sqrt{x} - 3\sqrt{y}} \right)^2$ is:

$$\frac{3x-2y}{2x+3y} = \frac{5}{6}$$

$$8x = 27y$$

$$\frac{x}{y} = \frac{27}{8}$$

$$\left(\frac{3\sqrt{27} + 3\sqrt{8}}{3\sqrt{27} - 3\sqrt{8}} \right)^2 \Rightarrow \left(\frac{3+2}{3-2} \right)^2$$

$$\Rightarrow \boxed{25} \text{ Ans}$$

94] If $a = b^p$, $b = c^q$, $c = a^r$ then pqr is

Soln:-

$$c = a^r = (b^p)^r$$

$$c = b^{pr} \Rightarrow (c^q)^{pr} \Rightarrow c^{pqr}$$

$$\boxed{pqr = 1} \text{ Ans}$$

95] The value of $(x^{\frac{1}{3}} + x^{-\frac{1}{3}})(x^{\frac{2}{3}} - 1 + x^{-\frac{2}{3}})$ is

Soln:- $a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$

$$\Rightarrow (x^{\frac{1}{3}} + x^{-\frac{1}{3}}) [(x^{\frac{1}{3}})^2 + (x^{-\frac{1}{3}})^2 - x^{\frac{1}{3}} \cdot x^{-\frac{1}{3}}]$$

$$\Rightarrow (x^{\frac{1}{3}})^3 + (x^{-\frac{1}{3}})^3$$

$$\Rightarrow x + x^{-1} \Rightarrow \boxed{x + \frac{1}{x}} \text{ Ans}$$

96] If $a = \frac{\sqrt{7} + \sqrt{6}}{\sqrt{7} - \sqrt{6}}$ and $b = \frac{\sqrt{7} - \sqrt{6}}{\sqrt{7} + \sqrt{6}}$, then find $\frac{a^2}{b} + \frac{b^2}{a} = ?$

Soln:- $b = \frac{1}{a}$

Put $\rightarrow a^3 + \frac{1}{a^3}$

$$a + \frac{1}{a} = \frac{\sqrt{7} + \sqrt{6}}{\sqrt{7} - \sqrt{6}} + \frac{\sqrt{7} - \sqrt{6}}{\sqrt{7} + \sqrt{6}}$$

$$= \frac{2(7+6)}{(7-6)} = 26$$

$$a^3 + \frac{1}{a^3} = (a + \frac{1}{a})^3 - 3(a + \frac{1}{a})$$

$$\Rightarrow (26)^3 - 3 \times 26$$

$$\Rightarrow \boxed{17498} \text{ Ans}$$

97] What is the value of $\left[\frac{1}{1-x^{(p-q)}} + \frac{1}{1-x^{(q-p)}} \right]$?

Soln:- $\left[\frac{1}{1-\frac{x^p}{x^q}} + \frac{1}{1-\frac{x^q}{x^p}} \right]$

$$\Rightarrow \left[\frac{x^q}{x^q-x^p} + \frac{x^p}{x^p-x^q} \right] \Rightarrow \frac{(x^q-x^p)}{(x^q-x^p)} = \boxed{1} \text{ Ans}$$

98] If $M = 0.1 + (0.1)^2 + (0.01)^2$ and $N = 0.3 + (0.03)^2 + (0.003)^2$, then what is the value of $M+N$?

Soln:- $M+N = 0.1 + (0.1)^2 + (0.01)^2 + 0.3 + (0.03)^2 + (0.003)^2$
 $\Rightarrow 0.1 + (0.1)^2 + (0.01)^2 + 0.3 + 9(0.01)^2 + 9(0.001)^2$
 $\Rightarrow 0.4 + 10(0.01)^2 + 9(0.001)^2 + (0.1)^2$
 $\Rightarrow 0.4 + 0.01 + 0.001 + 0.000009$
 $\Rightarrow \boxed{0.411009} \text{ Ans}$

99] What is the value of x in the equation

$$\sqrt{\frac{1+x}{x}} - \sqrt{\frac{x}{1+x}} = \frac{1}{\sqrt{6}} ?$$

(a) -2

(b) 3

(c) 2

(d) None of these

Soln: Option method:-

Put $x=2$ $\sqrt{\frac{3}{2}} - \sqrt{\frac{2}{3}}$

$$\frac{\sqrt{3}}{\sqrt{2}} - \frac{\sqrt{2}}{\sqrt{3}} \Rightarrow \frac{3-2}{\sqrt{6}} = \frac{1}{\sqrt{6}} \checkmark \text{ satisfy}$$

Order

100 Which among $2^{\frac{1}{2}}$, $3^{\frac{1}{3}}$, $4^{\frac{1}{4}}$, $6^{\frac{1}{6}}$ and $12^{\frac{1}{12}}$ is the largest

Soln :- $\text{Lcm}(2, 3, 4, 6, 12) = 12$

$$(2^{\frac{1}{2}})^{12}, (3^{\frac{1}{3}})^{12}, (4^{\frac{1}{4}})^{12}, (12^{\frac{1}{12}})^{12}$$

$$\Rightarrow 2^6, 3^4, 4^3, 12 \quad \rightarrow \text{Smallest} \Rightarrow 12^{\frac{1}{12}}$$

↓
largest = $3^{\frac{1}{3}}$

101 Which of the following relation is/are true?

(I) $(27)^{\frac{1}{3}} > (13)^{\frac{1}{2}} < (47)^{\frac{1}{6}}$

(II) $(23)^{\frac{1}{3}} < (49)^{\frac{1}{2}} < (52)^{\frac{1}{6}}$

(III) $(53)^{\frac{1}{6}} < (41)^{\frac{1}{3}} < (37)^{\frac{1}{2}}$

Soln :- (I) $\text{Lcm}(3, 2, 6) = 6$

$$27^2, 13^3, 47 \quad (\text{false})$$

$$529, 2197, 47$$

(II) $\text{Lcm} = 6$

$$23^2, 49^3, 52 \quad (\text{false})$$

$$(529) < (-) > 52$$

(III)

$$53, 41^2, 37^3$$

$$53 < 41^2 < 37^3 \quad \text{true} \checkmark$$

102 Which of the following statement is/are true?

(I) $(65)^{\frac{1}{6}} > (17)^{\frac{1}{4}} > (12)^{\frac{1}{3}}$

(II) $(17)^{\frac{1}{4}} > (65)^{\frac{1}{6}} > (12)^{\frac{1}{3}}$

~~(III)~~ $(12)^{\frac{1}{3}} > (17)^{\frac{1}{4}} > (65)^{\frac{1}{6}}$

Soln:- $(65^{\frac{1}{6}})^{12}, (17^{\frac{1}{4}})^{12}, (12^{\frac{1}{3}})^{12}$ $\text{Lcm}(6, 4, 3) = 12$

$\Rightarrow 65^2, 17^3, 12^4$

$\Rightarrow 4225 < 4913 < 20736$

103 Which value among $3^{200}, 2^{300}$ and 7^{100} is the largest?

Soln:- $3^{200}, 2^{300}, 7^{100}$

divide all power by 100 $\leftarrow \text{HCF}(200, 300, 100)$

$3^2, 2^3, 7^1$

9, 8, 7

$\uparrow$
greatest = $\boxed{3^{200}}$ Ans

104 The greatest number among $2^{60}, 3^{48}, 4^{36}$ and 5^{24} is:

Soln:- $2^{60}, 3^{48}, 4^{36}, 5^{24}$

$\text{HCF}(60, 48, 36, 24) = 12$

Power $\div 12$

$2^5, 3^4, 4^3, 5^2$

32, 81, 64, 25

$\uparrow$
greatest = $\boxed{3^{48}}$ Ans

105 Which one among $\sqrt[3]{6}$, $\sqrt[2]{5}$ and $\sqrt[4]{12}$ is the largest?

Soln:- $(6)^{\frac{1}{3}}$, $(5)^{\frac{1}{2}}$, $(12)^{\frac{1}{4}}$ $\text{Lcm}(3, 2, 4) = 6$

$$\Rightarrow 6^2, 5^3, 12^1$$

$$\Rightarrow 36, 125, 12$$

$\uparrow$
largest $\Rightarrow \boxed{\sqrt[2]{5}}$ Ans

106 Which value among $\sqrt[4]{7}$, $\sqrt[3]{11}$ and $\sqrt[12]{1257}$ is the largest

Soln:- $(7)^{\frac{1}{4}}$, $(11)^{\frac{1}{3}}$, $(1257)^{\frac{1}{12}}$ $\text{Lcm}(4, 3, 12) = 12$

Power $\times 12$

$$\Rightarrow 7^3, 11^4, 1257$$

$$\Rightarrow 343, (121)^2, 1257$$

$\uparrow$
greatest $\Rightarrow \boxed{\sqrt[3]{11}}$ Ans

107 Which of the following is true?

(I) $\sqrt[3]{11} > \sqrt{7} > \sqrt[4]{45}$

(II) $\sqrt{7} > \sqrt[3]{11} > \sqrt[4]{45}$

(III) $\sqrt{7} > \sqrt[4]{45} > \sqrt[3]{11}$

(IV) $\sqrt[4]{45} > \sqrt{7} > \sqrt[3]{11}$

Soln:- $(11)^{\frac{1}{3}}$, $(7)^{\frac{1}{2}}$, $(45)^{\frac{1}{4}}$ $\text{Lcm}(3, 2, 4) = 12$

Power $\times 12$

$$\Rightarrow 11^4, 7^6, 45^3$$

$$\Rightarrow 14641, 117649, 91125$$

$$\Rightarrow 117649 > 91125 > 14641$$

$$\sqrt{7} > \sqrt[4]{45} > \sqrt[3]{11} \quad \underline{\text{Ans}}$$

108 Which of the following is true?

(I) $\frac{1}{\sqrt[3]{12}} > \frac{1}{\sqrt[4]{29}} > \frac{1}{\sqrt{5}}$

(II) $\frac{1}{\sqrt[4]{29}} > \frac{1}{\sqrt[3]{12}} > \frac{1}{\sqrt{5}}$

(III) $\frac{1}{\sqrt{5}} > \frac{1}{\sqrt[3]{12}} > \frac{1}{\sqrt[4]{29}}$

(IV) $\frac{1}{\sqrt{5}} > \frac{1}{\sqrt[4]{29}} > \frac{1}{\sqrt[3]{12}}$

Soln:- $\sqrt[3]{12}$, $\sqrt[4]{29}$, $\sqrt{5}$

Take power = 12

$\text{LCM}(3, 4, 2) = 12$

12^4 , 29^3 , 5^6
 144^3 , $29^2 \times 29$, 125^2
 $\parallel$
 $(145+)^2$

If $a > b > c$
 then $\frac{1}{a} < \frac{1}{b} < \frac{1}{c}$

$\sqrt[4]{29} > \sqrt[3]{12} > \sqrt{5}$

then $\frac{1}{\sqrt[4]{29}} < \frac{1}{\sqrt[3]{12}} < \frac{1}{\sqrt{5}}$ Ans

109 The ascending order of $(2.89)^{0.5}$, $2 - (0.5)^2$, $\sqrt{3}$ and $\sqrt[3]{0.008}$ is

Soln:-
 $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 $\sqrt{2.89}$ $2 - 0.25$ $\downarrow$ $\downarrow$
 $\parallel$ $\parallel$ $\downarrow$ $\downarrow$
 1.7 1.75 1.732 0.2

Order:- $\sqrt[3]{0.008} < (2.89)^{0.5} < \sqrt{3} < 2 - (0.5)^2$ Ans

110 Which value among $\sqrt{11} + \sqrt{5}$, $\sqrt{14} + \sqrt{2}$, $\sqrt{8} + \sqrt{8}$ is the largest?

Soln:-

Type-1

If Sum $\Rightarrow$ Same
* then check:- their product

m.f $11+5 = 16$, $14+2 = 16$, $8+8 = 16$ (Same)

check:- product:

$11 \times 5 = 55$, $14 \times 2 = 28$, $8 \times 8 = 64$

greatest = 64 $\Rightarrow \boxed{\sqrt{8} + \sqrt{8}}$ Ans

m.f

$(\sqrt{11} + \sqrt{5})^2$, $(\sqrt{14} + \sqrt{2})^2$, $(\sqrt{8} + \sqrt{8})^2$
 $\Rightarrow 16 + 2\sqrt{55}$, $16 + 2\sqrt{28}$, $16 + 2\sqrt{64}$
 Greatest

111 The smallest in $(\sqrt{8} + \sqrt{5})$, $(\sqrt{7} + \sqrt{6})$, $(\sqrt{10} + \sqrt{3})$ and $(\sqrt{11} + \sqrt{2})$ is

Soln:-

Product:- 40, 35, 30, 22
 ↑
 Smallest
 $(\sqrt{11} + \sqrt{2})$ Ans

112 The greatest in $(\sqrt{19} + \sqrt{31})$, $(\sqrt{23} + 3\sqrt{3})$, $(\sqrt{17} + \sqrt{33})$, 10 ?

Soln

$(\sqrt{19} + \sqrt{31})$, $(\sqrt{23} + 3\sqrt{3})$, $(\sqrt{17} + \sqrt{33})$, $(\sqrt{25} + \sqrt{25})$

Product? 19×31 , 23×27 , 17×33 , 25×25
 diff $\Rightarrow 0$ greatest = 10 Ans

* $\left[\begin{array}{l} (x \times y)_{\max} \Rightarrow \text{difference of } x \text{ \& } y \text{ should be minimum,} \\ (x \times y)_{\min} \Rightarrow \text{difference of } x \text{ \& } y \text{ should be maximum,} \end{array} \right]$

113] Which of the following statement is true?

(I) $\sqrt{11} + \sqrt{7} < \sqrt{10} + \sqrt{8}$

(II) $\sqrt{17} + \sqrt{11} > \sqrt{15} + \sqrt{13}$

Soln:- $77 < 80$ (✓)

$187 > 195$ (X)

Sum $\Rightarrow$ Same

Sum $\Rightarrow$ Same

Type-2

(If sign are -ve) (Sum $\Rightarrow$ Same)

114] Which is the greatest among
 $(\sqrt{15} - \sqrt{10})$, $(\sqrt{19} - \sqrt{6})$, $(\sqrt{18} - \sqrt{7})$, $(\sqrt{17} - \sqrt{8})$?

Soln:- Basic:- Square it

$25 - 2\sqrt{150}$, $25 - 2\sqrt{114}$, $25 - 2\sqrt{126}$, $25 - 2\sqrt{136}$

$\uparrow$
small

So greatest = $(\sqrt{19} - \sqrt{6})$ Ans

Trick :- $\left\{ \begin{array}{l} \text{check product} \\ \text{If Product is maximum} \Rightarrow \text{No. will be minimum} \\ \text{If Product is minimum} \Rightarrow \text{No. will be maximum} \end{array} \right.$

Product 15×10 , 19×6 , 18×7 , 17×8
 $\Rightarrow$ 150, 114, 126, 136

$\uparrow$
minimum

So, No. will be maximum/greatest = $(\sqrt{19} - \sqrt{6})$ Ans

Type-3

Product $\Rightarrow$ Same

then check:- Sum

if Sum_{max} = Largest

Sum_{min} = Smallest

115 Which is the greatest among
 $(\sqrt{24} + \sqrt{10})$, $(\sqrt{30} + \sqrt{8})$, $(\sqrt{15} + \sqrt{16})$, $(\sqrt{12} + \sqrt{20})$?

Soln:- Product = 240 (same)

Sum:- 34 , 38 , 31 , 32

↑
largest $\Rightarrow (\sqrt{30} + \sqrt{8})$ Ans

116 Which is the smallest among.

$\sqrt{10} - \sqrt{4}$, $\sqrt{20} - \sqrt{2}$, $\sqrt{8} - \sqrt{5}$

Soln

$\sqrt{+}$
14

$\sqrt{+}$
22

$\sqrt{+}$
13

↑
Smallest

$\sqrt{8} - \sqrt{5}$ Ans

Type-4

(Difference is same)

117 Which is the greatest among
 $(\sqrt{17} - \sqrt{14})$, $(\sqrt{19} - \sqrt{16})$, $(\sqrt{22} - \sqrt{19})$, $(\sqrt{13} - \sqrt{10})$

Soln:-

$$\frac{3}{\sqrt{17} + \sqrt{14}}$$

+
31

$$\frac{3}{\sqrt{19} + \sqrt{16}}$$

+
35

$$\frac{3}{\sqrt{22} + \sqrt{19}}$$

+
41

$$\frac{3}{\sqrt{13} + \sqrt{10}}$$

+
23

↑
Smallest denominator

⇒ Greatest $(\sqrt{13} - \sqrt{10})$ Ans

118 Which one among the following is the smallest?

(a) $\sqrt{201} - \sqrt{199}$

(b) $\sqrt{101} - \sqrt{99}$

(c) $\sqrt{301} - \sqrt{299}$

(d) $\sqrt{401} - \sqrt{399}$

Soln:-

$$\frac{+}{400}$$

$$\frac{+}{200}$$

$$\frac{+}{600}$$

$$\frac{+}{800}$$

↑
min sum,

↓
So Greatest

$$\boxed{\sqrt{101} - \sqrt{99}}$$

↑
max sum

So **Smallest**

$$\boxed{\sqrt{401} - \sqrt{399}}$$

Ans

119) Which of the following relation is true?

(I) $3^{33} > 33^3$ (II) $33^3 > 333$ (III) $3^{33} > 333$

Soln:-

(I) $\rightarrow 3^{33} > 33^3$

$(3^3)^{11} \Rightarrow (27)^{11} > 33^3 \quad (\checkmark)$

(II) $\rightarrow 33^3 > 333 \quad (\checkmark)$

(III) $\rightarrow 3^{33} > 333 \quad (\checkmark)$

120) Which of the following statement is true.

(I) $\frac{3}{71} < \frac{5}{91} < \frac{7}{99}$

(II) $\frac{11}{135} > \frac{12}{157} > \frac{13}{181}$

Soln:-

(I) $\rightarrow \frac{3}{71} \times \frac{5}{91}$
 $273 < 355$

$\Rightarrow \frac{3}{71} < \frac{5}{91}$

Now

$\frac{5}{91} \times \frac{7}{99}$
 $495 < 637$

$\frac{5}{91} < \frac{7}{99}$

(✓)

(II) $\frac{11}{135} \times \frac{12}{157}$

$1727 > 1620$

$\frac{11}{135} > \frac{12}{157}$

$\frac{12}{157} \times \frac{13}{181}$

$12 \times 181 > 13 \times 157$

$\frac{12}{157} > \frac{13}{181}$

$\frac{11}{135} > \frac{12}{157} > \frac{13}{181} \quad (\checkmark)$

121 Which of the following statement is true.

(I) $\sqrt{121} + \sqrt{12321} + \sqrt{1234321} = 1233$

(II) $\sqrt{0.64} + \sqrt{64} + \sqrt{36} + \sqrt{0.36} > 15$

Soln:- (I) $\sqrt{121} + \sqrt{12321} + \sqrt{1234321}$
 $\Rightarrow 11 + 111 + 1111$
 $\Rightarrow 1233 (\checkmark)$

(II) $\sqrt{0.64} + \sqrt{64} + \sqrt{36} + \sqrt{0.36}$
 $\Rightarrow 0.8 + 8 + 6 + 0.6$
 $\Rightarrow 15.4 > 15 (\checkmark)$

122 If $a^x = \sqrt{b}$, $b^y = \sqrt[3]{c}$, $c^z = \sqrt{a}$, then find $xyz = ?$

Soln:- $a^x = \sqrt{b}$ | $b^y = \sqrt[3]{c}$ | $c^z = \sqrt{a}$
 $b = a^{2x}$ | $(a^{2x})^y = (a^{\frac{1}{2z}})^{\frac{1}{3}}$ | $c = a^{\frac{1}{2z}}$
 $a^{2xy} = a^{\frac{1}{6z}}$

$$2xy = \frac{1}{6z}$$

$$\boxed{xyz = \frac{1}{12}}$$

Ans